CHAPTER 1
INTRODUCTION

1.1 Sets and subsets

1.1.1 Sets

- Any well-defined collection of objects called the *elements* or *members of the set.*

  Example 1: {1, 2, 3} – set of all +ve integer < 4

- The order in which the elements of a set are listed is not important, thus:

  {1, 2, 3}, {2, 1, 3}, {3, 2, 1} – all represent the set given in *example 1*

- Repeated elements in the set can be ignored, thus:

  {1, 2, 3, 3, 2, 1} – represent the set given in *example 1*

- Uppercase letters: *A, B, C* … – denote sets
  Lowercase letters: *a, b, c* … – denotes the elements of sets

  $x \in A$: x is an element of the set A

  $x \notin A$: x is not an element of set A

  Example 2: $A = \{1, 3, 5, 7\}$

  Then, $1 \in A$, $3 \in A$, but $4 \notin A$

- Another useful way to define a set is by specifying a property that the elements of the set have in common – if the elements is too many

Example 3: $\{1, 2, 3\} \rightarrow \{x \mid x \text{ is a +ve integer} < 4\}$

$\{b, y, t, e\} \rightarrow \{x \mid x \text{ is a letter in the word "byte"}\}$

- *Definition 1*: some of the common sets of numbers and the notations

i) $N = \{x \mid x \text{ is the +ve integer and zero}\}$ e.g. $\{0, 1, 2, 3 \ldots\}$
ii) $Z = \{x \mid x \text{ is the set of integers}\}$ e.g. $\{\ldots -1, 0, 1, 2 \ldots\}$
iii) $Z^+ = \{x \mid x \text{ is the set of +ve integers}\}$ e.g. $\{1, 2, 3 \ldots\}$
iv) $Z^{\geq 0}$ is the same as N
v) $R = \{x \mid x \text{ is the set of real numbers}\}$ e.g. $\{1.12, 4.9 \ldots\}$
vi) $Q = \{x \mid x \text{ is the set of rational numbers}\}$

Thus Q consists of numbers that can be written as a/b, where a and b are integers and $b \neq 0$.

vii) $\varnothing$ = empty set or { } or set with no element

Example 4: $A = \{x \mid x \text{ is a real number an } x^2 = -1\}$

Then, $A = \varnothing$

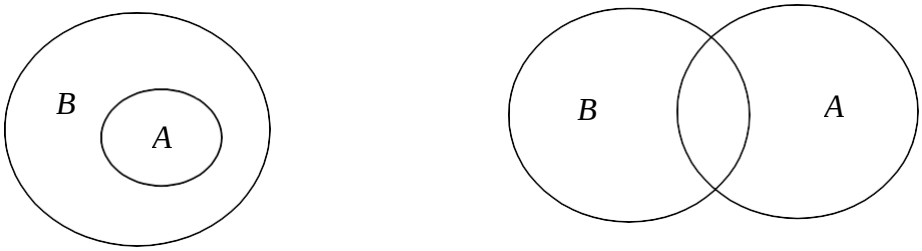
- $A = B$ or A and B are equal if they have the same elements

Example 5: $A = \{1, 2, 3\}$ and $B = \{x \mid x \text{ is a +ve integer and } x^2 < 12\}$
Then, $A = B$

Example 6: $A = \{\text{JAVA, C, ADA}\}$ and $B = \{\text{ADA, JAVA, C}\}$
Then, $A = B$

1.1.2 Subsets

- A is a subset of B or that A is contained in B or $A \subseteq B$
- A is not a subset of B or $A \nsubseteq B$
- Figure 1.1 below, which are used to show relationships between sets are called *Venn diagrams.*



$A \subseteq B$ $A \nsubseteq B$

Figure 1.1

Example 7: $Z^+ \subseteq Z$

Example 8: Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{2, 4, 5\}$ and $C = \{1, 2, 3, 4, 5\}$

Then, $B \subseteq A$, $B \subseteq C$ and $C \subseteq A$

However, $A \nsubseteq B$, $A \nsubseteq C$ and $C \nsubseteq B$

Example 9: If A is any set, then $A \subseteq A$

Example 10: Let A be a set and let $B = \{A, \{A\}\}$

Then, $A \in B$ and $\{A\} \in B$

$\{A\} \subseteq B$ and $\{\{A\}\} \subseteq B$

However, $A \notin B$

Example 11: For any set A, $\varnothing \subseteq B$

$A = B$ if and only if $A \subseteq B$ and $B \subseteq A$

- Set U (universal set) – set contained all object.
  Any other set is assumed to be a subset of U.

- Figure 1.2 shows the Venn diagrams for the universal set U that are denoted by a rectangle, while sets within U are denoted by circles

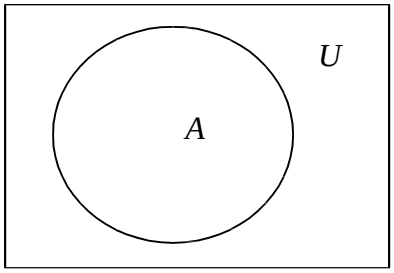


Figure 1.2

- Finite set – set with n distinct elements, where $n \in N$
  n is called the cardinality of A and is denoted as $|A|$

- Infinite set – set that is not finite

  The sets introduced in *Definition 1* (except $\varnothing$) are infinite sets

- Power set – if A is a set, then the set of all subsets is called the power set of A and is denoted by $P(A)$.

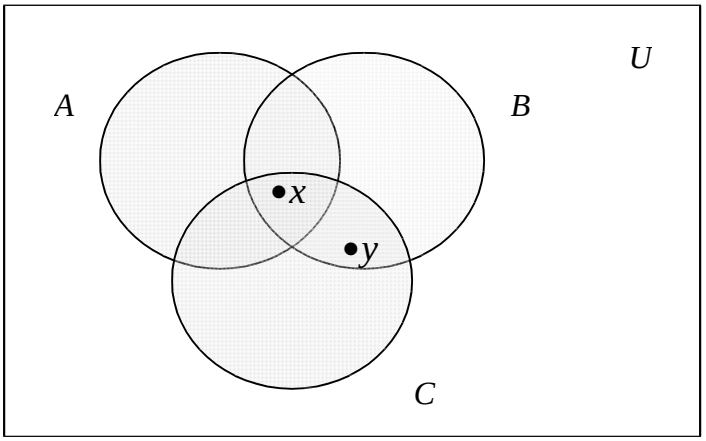
Example 12: Let $A = \{1, 2, 3\}$
$P(A) = \{\{\ \}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$
$|A| = 3$
$|P(A)| = 8$

Example 13: If $P(B) = \{\{a\}, \{\ \}, \{c\}, \{b, c\}, \{a, b\}, \ldots\}$ and $|P(B)| = 8$, find B.

Answer:
$B = \{a, b, c\}$

Example 14: Let $x \in A$, $x \in B$, $x \in C$, $y \in B$, $y \in C$ and $y \notin A$. Draw a Venn diagrams.

Answer:



1.2 Operations on Sets

❖ Union (*kesatuan*)

- If A and B are sets, we define their union as the set consisting of all elements that belong to A or B and denote it by $A \cup B$

- $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$

Example 1: Let $A = \{a, b, c, e, f\}$ and $B = \{b, d, r, s\}$
Then, $A \cup B = \{a, b, c, d, e, f, r, s\}$

- Figure 1.3 (a) illustrated the union of two sets with a Venn diagram

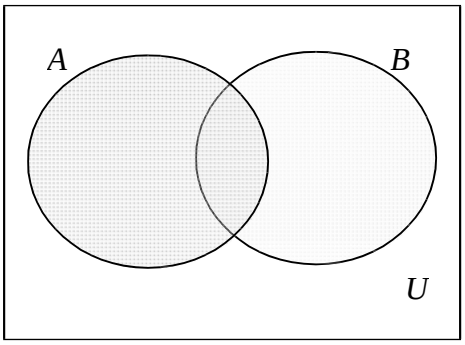
❖ Intersection (*persilangan*)

- If A and B are sets, we define their intersection as the set consisting of all elements that belong to both A and B and denote it by $A \cap B$

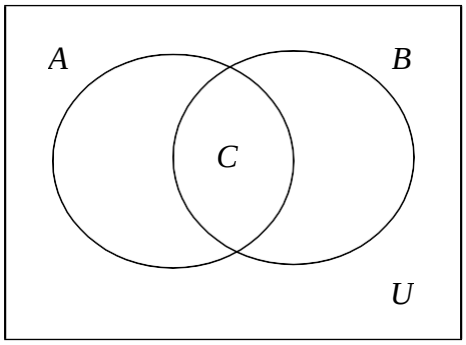
- $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

Example 2: Let $A = \{a, b, c, e, f\}$, $B = \{b, e, f, r, s\}$ and $C = \{a, t, u, v\}$
Then, $A \cap B = \{b, e, f\}$
$A \cap C = \{a\}$
$B \cap C = \varnothing$

- Figure 1.3 (b) illustrated the intersection of two sets with a Venn diagram



(a) $A \cup B$



(b) $C = A \cap B$

Figure 1.3

❖ Disjoint sets (*set tak bercantum*)

- Two sets that have no common elements

- Figure 1.4 illustrated the two disjoint sets using a Venn diagram

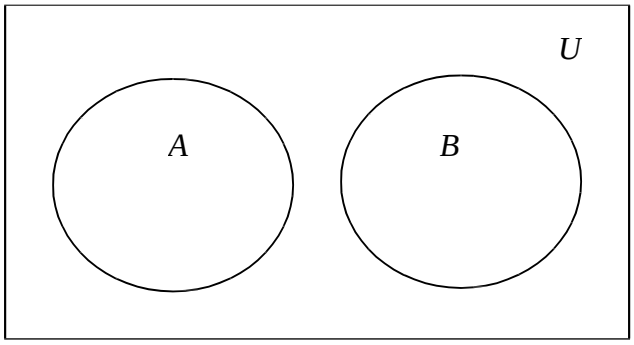


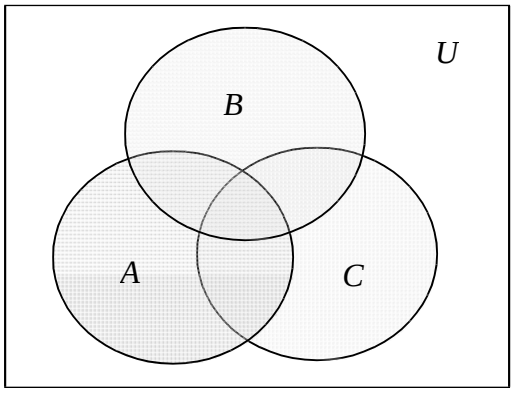
Figure 1.4: two disjoint sets

- The operations of union and intersection can be defined for three or more sets in an obvious manner as follows and as illustrated in figure 1.5

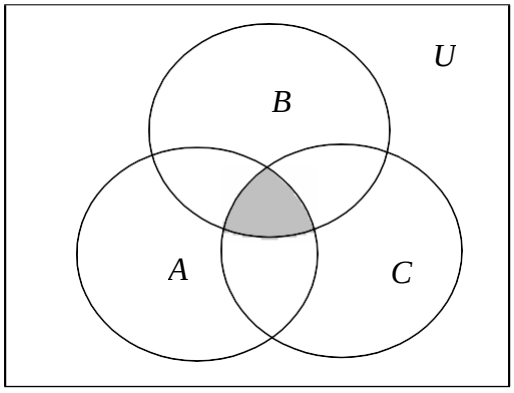
  $A \cup B \cup C = \{x \mid x \in A \text{ or } x \in B \text{ or } x \in C\}$

  and;

  $A \cap B \cap C = \{x \mid x \in A \text{ and } x \in B \text{ and } x \in C\}$



$A \cup B \cup C$



$A \cap B \cap C$

Figure 1.5

Example 3: Let $A = \{1, 2, 3, 4, 5, 7\}$, $B = \{1, 3, 8, 9\}$ and $C = \{1, 3, 6, 8\}$

Then, $A \cap B \cap C = \{1, 3\}$

❖ Complement of B with respect to A (*set pelengkap*)

- If A and B are two sets, we define the complement of B with respect to A as the set of all elements that belong to A but not to B

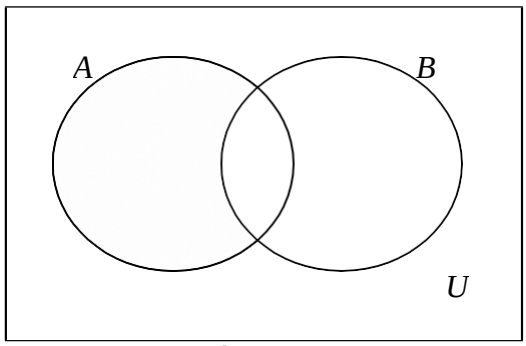
- Denote it by: $A - B$,

  Thus: $A - B = \{x \mid x \in A \text{ and } x \notin B\}$

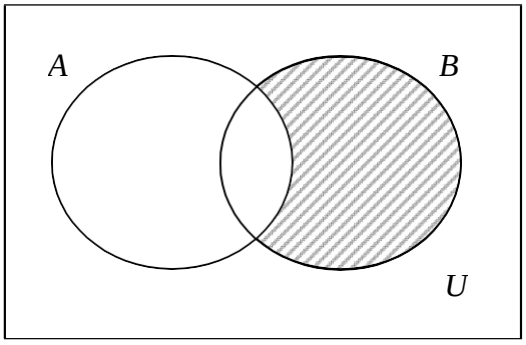
Example 4: Let $A = \{a, b, c\}$ and $B = \{b, c, d, e\}$

Then, $A - B = \{a\}$ and $B - A = \{d, e\}$

- As illustrated in figure 1.6



$A - B$



$B - A$

Figure 1.6

- If U is a universal set containing A, then $U - A$ is called the complement of A (*set pelengkap A*) and is denoted by $\bar{A}$ as illustrated in figure 1.7

  Thus: $\bar{A} = \{x \mid x \notin A\}$

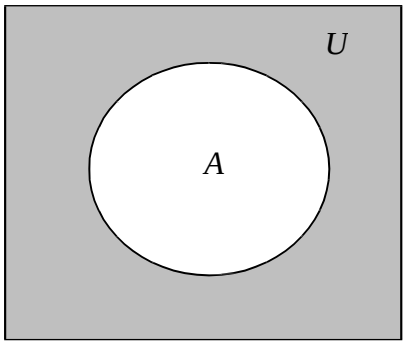


Figure 1.7

❖ Symmetric difference (*beza simetri*)

- If A and B are two sets, we define their symmetric difference as the set of all elements that belong to A or B, but not to both A and B

- Denote it by: $A \oplus B$

- Thus: $A \oplus B = \{x \mid (x \in A \text{ and } x \notin B) \text{ or } (x \in B \text{ and } x \notin A)\}$

  $A \oplus B = (A - B) \cup (B - A)$

Example 5: Let $A = \{a, b, c, d\}$ and $B = \{a, c, e, f, g\}$

Then, $A \oplus B = \{b, d, e, f, g\}$

- Symmetric difference is illustrated in figure 1.8

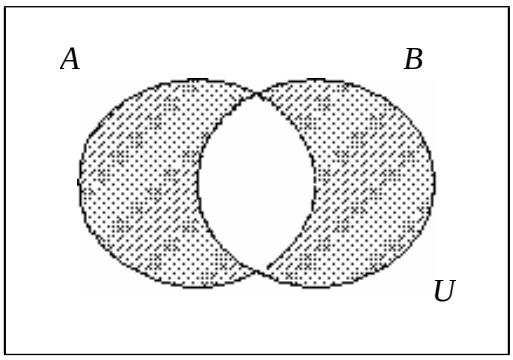


Figure 1.8

- ***Theorem 1*** – The operations defined on sets satisfy the following properties:

Commutative Properties (Sifat-sifat kalis tukar tertib)

1. $A \cup B = B \cup A$
2. $A \cap B = B \cap A$

Associate Properties (Sifat-sifat kalis sekutuan)

3. $A \cup (B \cup C) = (A \cup B) \cup C$
4. $A \cap (B \cap C) = (A \cap B) \cap C$

Distributive Properties (Sifat-sifat kalis agihan)

5. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
6. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Idempotent Properties (Sifat-sifat idempotent)

7. $A \cup A = A$

8. $A \cap A = A$

Properties of the Complement (Sifat-sifat pelengkap)

9. $\overline{\overline{(A)}} = A$

10. $A \cup \bar{A} = U$

11. $A \cap \bar{A} = \varnothing$

12. $\overline{\varnothing} = U$

13. $\overline{U} = \varnothing$

14. $\overline{A \cup B} = \overline{A} \cap \overline{B}$

15. $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Properties 14 and 15 are known as De Morgan's laws

Properties of the Universal Set (Sifat-sifat set sejagat)

16. $A \cup U = U$

17. $A \cap U = A$

Properties of the Empty Set (Sifat-sifat set kosong)

18. $A \cup \varnothing = A$ or $A \cup \{\ \} = A$

19. $A \cap \varnothing = \varnothing$ or $A \cap \{\ \} = \{\ \}$

- ***Theorem 2*** – if A and B are finite sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example 6: Let $A = \{a, b, c, d, e\}$ and $B = \{c, e, f, h, k, m\}$

Then, $A \cup B = \{a, b, c, d, e, f, h, k, m\}$

And, $A \cap B = \{c, e\}$

Therefore, $|A| = 5$

$|B| = 6$

$|A \cup B| = 9$

$|A \cap B| = 2$

Hence, $|A \cup B| = |A| + |B| - |A \cap B|$

$|A \cup B| = 5 + 6 - 2$

$|A \cup B| = 9$

Example 7: If A and B are disjoint sets or $A \cap B = \varnothing$, therefore

$$|A \cup B| = |A| + |B|$$

- ***Theorem 3*** – Let A, B and C be finite sets, then

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Example 8: Let $A = \{a, b, c, d, e\}$, $B = \{a, b, e, g, h\}$ and $C = \{b, d, e, g, h, k, m, n\}$

Then,
$$|A| = 5$$
$$|B| = 5$$
$$|C| = 8$$
$$|A \cap B| = |\{a, c, e\}| = 3$$
$$|B \cap C| = |\{b, e, g, h\}| = 4$$
$$|A \cap C| = |\{b, d, e\}| = 3$$
$$|A \cap B \cap C| = |\{b, e\}| = 2$$

Therefore,
$$|A \cup B \cup C| = 5 + 5 + 8 - 3 - 4 - 3 + 2$$
$$|A \cup B \cup C| = 10$$

Example 9: If A, B and C are disjoint sets or $A \cap B \cap C = \varnothing$, therefore

$$|A \cup B \cup C| = |A| + |B| + |C|$$

1.3 Sequence

- A list of objects arranged in a definite order; a first element, second element, third element, and so on
- Can be finite or infinite
- The element may be different, or repeated

Example 1: 1, 0, 0, 1, 0, 1, 0, 0, 1, 1 → finite sequence with repeated items

Example 2: 3, 8, 13, 18, 23, … → an infinite sequence

Example 3: 1, 4, 9, 16, 25, … → an infinite sequence of all +ve integers squares

- Sequence can be written as a formula

❖ Recursive formula (*rumus rekursif*)
- A formula that refers to previous term to define the next term
- Must include a starting place
- This type of formula is called *explicit*

Example 4: 3, 8, 13, 18, 23, … can be written as

$$a_1 = 3,\ a_n = a_{n-1} + 5,\ 2 \leq n \leq \infty$$

Example 5: 1, 4, 9, 16, 25, … can be written as

$$b_n = n^2,\ 1 \leq n \leq \infty$$

❖ String (*rentetan*)

- Sequences of letters or other symbols

  Example 6: C, O, M, P, U, T, E, R → in a sequence of 'COMPUTER'

- It can be finite or infinite

  Example 7: Sequence 'abababab…' → infinite sequence a, b, a, b…

  Sequence 'welcome to this class' → finite sequence

- The set corresponding to a sequence is simply the set of all distinct elements in the sequence

  Example 8: Infinite sequence 1, 4, 9, 16, 25, …
  is {1, 4, 9, 16, 25, …} (in a form of set)

  Infinite sequence abababab…
  is simply {a, b} (in a form of set)

❖ Sequence and Array (*jujukan dan tatasusunan*)

- The idea of a sequence is important in computer science

- Sequence is represented in any programming language as an array

  Example 9: infinite sequence *S*: 0, 1, 2, 3, 2, 1,…

  An array is viewed as a sequence of positions as present below:

	S[1]	*S*[2]	*S*[3]	…
Array *S*:	0	1	2	…
	1	2	3	…

1.3.1 Characteristic Function (*fungsi cirian*)

- A function on a set as a rule that assigns some value to each elements of the set

Example 10: If $A \subseteq U$, the characteristic function f_A of A is defined for each $x \in U$ as follows:

$$f_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

Example 11: If $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2\}$, $B = \{2, 4, 6\}$ and $C = \{4, 5, 6\}$, find f_A?

$$f_A(x) = \begin{cases} 1 & \text{if } x = 1 \text{ or } 2 \\ 0 & \text{if others} \end{cases}$$

Hence, f_A corresponds to the sequence 1, 1, 0, 0, 0, 0.

Find f_B and f_C.

1.4 Properties of Integers

- If m is an integer and n is a positive integer, we can plot the integer multiples of n on a line, and locate m as in Figure 1.9.

- If m is a multiple of n, say $m = qn$, then we can write $m = qn + r$, where r is 0.

- If m is not a multiple of n, we let qn be the first multiple of n lying to the left of m and let r be $m - qn$.

- Then r is the distance from qn to m, so clearly $0 < r < n$, and again we have $m = qn + r$.

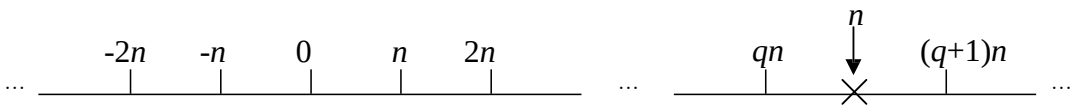


Figure 1.9

- ***Theorem 1*** – If n and m are integers and $n > 0$, we can write $m = qn + r$ for integers q and r with $0 \leq r < n$.

Example 1: a) If n is 3 and m is 16,
then $16 = 5(3) + 1$ so q is 5 and r is 1.
i) If n is 10 and m is 3,
then $3 = 0(10) + 3$ so q is 0 and r is 3.
ii) If n is 5 and m is -11,
then $-11 = -3(5) + 4$ so q is -3 and r is 4.

- If the r in Theorem 1 is zero, so that m is a multiple of n, we write $n \mid m$

- If $n \mid m$, then $m = qn$ and $n \leq |m|$

- If m is not a multiple of n, we write $n \nmid m$

- ***Theorem 2*** – Let a, b and c be integers
  a) If $a \mid b$ and $a \mid c$, then $a \mid (b + c)$
  b) If $a \mid b$ and $a \mid c$, where $b > c$, then a $\mid (b - c)$
  c) If $a \mid b$ or $a \mid c$, then $a \mid bc$
  d) If $a \mid b$ and $b \mid c$, then $a \mid c$

- A number $p > 1$ in Z^+ is called *prime* if the only positive integers that divide p are p and 1

  Example 2: The numbers 2, 3, 5, 7, 11 and 13 are prime, while 4, 10, 16 and 21 are not prime.

- ***Theorem 3*** – Every positive integer $n > 1$ can be written uniquely as $p_1^{k1} p_2^{k2} \ldots p_s^{ks}$, *where* $p_1 < p_2 < \ldots < p_s$ are distinct primes that divide n and the k's are positive integers giving the number of times each prime occurs as a factor of n.

  Example 3: (a) $9 = 3 \cdot 3 = 3^2$
  (b) $24 = 8 \cdot 3 = 2 \cdot 2 \cdot 2 \cdot 3 = 2^3 \cdot 3$
  (c) $30 = 2 \cdot 3 \cdot 5$

1.4.1 Greatest Common Divisor (GCD)

- If a, b and k are in Z^+, and $k \mid a$ and $k \mid b$, we say that k is a *common divisor* of a and b.

- If d is the largest such k, d is called the *greatest common divisor*, of a and b. Denoted by $d = \text{GCD}(a, b)$

- ***Theorem 4*** – If d is GCD(a, b), then
  a) $d = sa + tb$ for some integers s and t
  b) If c is any other common divisor of a and b, then $c \mid d$

- Let a, b and d be in Z^+. The integer d is the GCD of a and b if and only if

  a) $d \mid a$ and $d \mid b$
  b) Whenever $c \mid a$ and $c \mid b$, then $c \mid d$.

Example 4: (a) The common divisors of 12 and 30 are 1, 2, 3, and 6, so that

GCD (12, 30) = 6 and 6 = 1 . 30 + (-2) . 12

(b) It is clearly that GCD (17, 95) = 1 since 17 is prime and 17 $\nmid$ 95, and the reader may verify that 1 = 28 . 17 + (-5) . 95

- If GCD (a, b) = 1, we say a and b are *relatively prime*

❖ Euclidean algorithm

- Is a procedure for finding GCD (a, b)

Example 5: Let a be 190 and b be 34. Find GCD using the Euclidean algorithm.

Answer:

Divide 190 by 34: 190 = 5 . 34 + 20
Divide 34 by 20: 34 = 1 . 20 + 14
Divide 20 by 14: 20 = 1 . 14 + 6
Divide 14 by 6: 14 = 2 . 6 + 2
Divide 6 by 2: 6 = 3 . 2 + 0

So, GCD (190, 34) = 2, the last of the nonzero divisors.

- If d = GCD (a, b), we can find integers s and t such that $d = sa + tb$.

Example 6: a) Let $a = 190$ and $b = 34$ as in Example 5. Then:

$$\begin{aligned}
\text{GCD } (190, 34) &= 2 \\
&= 14 - 2(6) \\
&= 14 - 2[20 - 1(14)] \\
&= 14 - 2(20) + 2(14) \\
&= 3(14) - 2(20) \\
&= 3[34 - 1(20)] - 2(20) \\
&= 3(34) - 3(20) - 2(20) \\
&= 3(34) - 5(20) \\
&= 3(34) - 5(190 - 5 \,.\, 34) \\
&= 3(34) - 5(190) + 25(34) \\
&= 28(34) - 5(190)
\end{aligned}$$

Hence, $s = -5$ and $t = 28$.

b) Let $a = 108$ and $b = 60$. Find s and t.

Answer:

$$\begin{aligned}
\text{GCD } (108, 60) &= 12 \\
&= 60 - 1(48) \\
&= 60 - 1[108 - 1(60)] \\
&= 60 - 108 + 60 \\
&= 2(60) - 108
\end{aligned}$$

Hence, $s = -1$ and $t = 2$.

- ***Theorem 5*** – If a and b are in Z^+, $b > a$, then GCD(a, b) = GCD(b, b, $\pm a$)

1.4.2 Least Common Multiple (LCM)

- If a, b and k are in Z^+, and $a \mid k$ and $b \mid k$, we say that k is a *common multiple* of a and b.

- The smallest such k, call it c, is called the *least common multiple*, LCM of a and b, denoted by c = LCM (a, b)

- ***Theorem 6*** – If a and b are two positive integers, then GCD (a, b) . LCM (a, b) = ab.

Example 7: Let $a = 540$ and $b = 504$. Factoring a and b into primes, we obtain

$a = 540 = 2^2 \cdot 3^3 \cdot 5$ and $b = 504 = 2^3 \cdot 3^2 \cdot 7$

Then, $a = 2^2 \cdot 3^3 \cdot 5^1 \cdot 7^0$ and $b = 2^3 \cdot 3^2 \cdot 5^0 \cdot 7^1$

$$\begin{aligned} \text{GCD } (540, 504) &= 2^{\min(2,3)} \cdot 3^{\min(2,3)} \cdot 5^{\min(1,0)} \cdot 7^{\min(1,0)} \\ &= 2^2 \cdot 3^2 \cdot 5^0 \cdot 7^0 \\ &= 2^2 \cdot 3^2 \text{ or } 36 \end{aligned}$$

Also,

$$\begin{aligned} \text{LCM } (540, 504) &= 2^{\max(2,3)} \cdot 3^{\max(2,3)} \cdot 5^{\max(1,0)} \cdot 7^{\max(1,0)} \\ &= 2^3 \cdot 3^3 \cdot 5^1 \cdot 7^1 \text{ or } 7560 \end{aligned}$$

Then,

$$\begin{aligned} \text{GCD } (540, 504) \cdot \text{LCM } (540, 504) &= 36 \cdot 7560 \\ &= 272160 \\ &= 540 \cdot 504 \end{aligned}$$

Example 8: Find the LCM of the given integers.

a) 150, 70

Answer:

Factoring a and b into primes, we obtain

$a = 150 = 2^1 \cdot 3^1 \cdot 5^2$ and $b = 70 = 2^1 \cdot 5^1 \cdot 7^1$

Then, $a = 2^1 \cdot 3^1 \cdot 5^2 \cdot 7^0$ and $b = 2^1 \cdot 3^0 \cdot 5^1 \cdot 7^1$

LCM (150, 70) $= 2^{\max(1,1)} \cdot 3^{\max(1,0)} \cdot 5^{\max(2,1)} \cdot 7^{\max(0,1)}$
$= 2^1 \cdot 3^1 \cdot 5^2 \cdot 7^1$
$= 1050$

b) 32, 27

Answer:

$= 864$

1.5 Matrices

- A rectangular array of numbers arranged in *m* horizontal rows and *n* vertical columns

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{1n} \\ . & . & & . \\ . & . & & . \\ . & . & & . \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

m (↓)

n (→)

- We say that A is *m* by *n*, written *m* x *n*.
- Often write as A = [a_{ij}]
- If *m* = *n* → a square matrix

Example 1: Let $A = \begin{bmatrix} 2 & 3 & 5 \\ 0 & -1 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix}$

$C = \begin{bmatrix} 1 & -1 & 3 & 4 \end{bmatrix}$ $D = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$

$E = \begin{bmatrix} -1 & 0 & -1 \\ -1 & 2 & 3 \\ -2 & 4 & 5 \end{bmatrix}$

Then, A = 2 x 3, B = 2 x 2, C = 1 x 4, D = 3 x 1 and E = 3 x 3

- A square matrix, A = [a_{ij}] for which every entry off the main diagonal is zero, that is $a_{ij} = 0$ for $i \neq j$, is called diagonal matrix (*matriks pepenjuru*)

Example 2: $F = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$ $G = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

$H = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 6 \end{bmatrix}$

F, G and H are a diagonal matrix.

- Two m x n matrices A = [a_{ij}] and B = [b_{ij}] are said to be equal if $a_{ij} = b_{ij}$, $1 \leq i \leq m$, $1 \leq j \leq n$;

Example 3: $A = \begin{bmatrix} 2 & 3 & 0 \\ 0 & -3 & 8 \\ -2 & 0 & 5 \end{bmatrix}$ $B = \begin{bmatrix} x & 3 & 0 \\ 0 & -3 & z \\ y & 0 & 5 \end{bmatrix}$

A = B if and only if $x = 2$, $y = -2$ and $z = 8$

- A matrix all of whose entries are zero is called a zero matrix (*matriks sifar*)

Example 4: $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

❖ Arithmetic operation on matrix

1. Addition
   - If A = [a_{ij}] and B = [b_{ij}] are m x n matrices (the number of rows and columns are same), then the sum of A and B is the matrix of C = [c_{ij}] defined by $c_{ij} = a_{ij} + b_{ij}$, $1 \leq i \leq m$, $1 \leq j \leq n$

   - The sum of two matrices A and B is defined only when A and B have the same number of rows and the same number of columns

Example 5: Let $A = \begin{bmatrix} 3 & 4 & -1 \\ 5 & 0 & -2 \end{bmatrix}$ $B = \begin{bmatrix} 4 & 5 & 3 \\ 0 & -3 & 2 \end{bmatrix}$

Then, $A + B = C = \begin{bmatrix} 7 & 9 & 2 \\ 5 & -3 & 0 \end{bmatrix}$

   - ***Theorem 1*** – basic properties of matrix addition

     i) $A + B = B + A$
     ii) $(A + B) + C = A + (B + C)$
     iii) $A + 0 = 0 + A = A$

2. Multiplication
   - If A = [a_{ij}] is an m x p matrix and B = [b_{ij}] is a p x n matrix, then the multiply value of A and B, denoted AB, is the m x n matrix C = [c_{ij}]

   - The number of matrix A column must be equal with matrix B row

Example 6: Let $A = \begin{bmatrix} 2 & 3 & -4 \\ 1 & 2 & 3 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 1 \\ -2 & 2 \\ 5 & -3 \end{bmatrix}$

Then, $AB = \begin{bmatrix} (2)(3) + (3)(-2) + (-4)(5) & (2)(1) + (3)(2) + (-4)(-3) \\ (1)(3) + (2)(-2) + (3)(5) & (1)(1) + (2)(2) + (3)(-3) \end{bmatrix}$

$= \begin{bmatrix} -20 & 20 \\ 14 & -4 \end{bmatrix}$

Example 7: Let $A = \begin{bmatrix} 2 & 1 \\ 3 & -2 \end{bmatrix}$ $B = \begin{bmatrix} 1 & -1 \\ 2 & -3 \end{bmatrix}$

Then, AB =

And BA =

Is AB = BA?

- ***Theorem 2*** – basic properties of matrix multiplication

  i) A (BC) = (AB) C
  ii) A (B + C) = AB + AC
  iii) (A + B) C = AC + BC

❖ Identity matrix

- All of whose diagonal elements are 1 for square matrix (n x n matrix)

Example 8: $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $I_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ . & . & & . \\ . & . & & . \\ . & . & & . \\ 0 & 0 & & 1 \end{bmatrix}$

- Operation on identity matrix includes:

ii) If A is an m x n matrix,

Then, $I_m A = A I_n = A$

iii) If A is n x n matrix and p is a +ve integer,

Then, $A^p = \underbrace{A \,.\, A \,\dots\, A}_{p \text{ factors}}$ and $A^0 = I_n$.

iv) If p and q are nonnegative integers,

Then, $A^p \, A^q = A^{p+q}$ and $(A^p)^q = A^{pq}$

❖ Transposition matrix

- If A = [a_{ij}] is an m x n matrix, then the n x m matrix A^T = [a^T_{ij}], where $a^T_{ij} = a_{ji}$, $1 \le i \le m$, $1 \le j \le n$

- The transpose of A is obtained by interchanging the rows and columns of A

Example 9: Let $A = \begin{bmatrix} 2 & -3 & 5 \\ 6 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 4 & 5 \\ 2 & -1 & 0 \\ 1 & 6 & 2 \end{bmatrix}$

Then, $A^T = \begin{bmatrix} 2 & 6 \\ -3 & 1 \\ 5 & 3 \end{bmatrix}$

And, $B^T = \begin{bmatrix} 3 & 2 & 1 \\ 4 & -1 & 6 \\ 5 & 0 & 2 \end{bmatrix}$

- ***Theorem 3*** – basic properties of the transpose operation

ii) $(A^T)^T = A$

iii) $(A + B)^T = A^T + B^T$

iv) $(AB)^T = A^T B^T$

❖ Symmetric

- A matrix A = [a_{ij}] is called symmetric if $A^T = A$
- If A is symmetric, it must be a square matrix
- A is symmetric if and only if the entries of A are symmetric with respect to the main diagonal of A

$$a_{ij} = a_{ji}$$

Example 10: If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & 5 \\ -3 & 5 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & 0 \\ 3 & 2 & 1 \end{bmatrix}$

Then, A is symmetric and B is not symmetric.

❖ Boolean matrix

- An *m* x *n* matrix whose entries are either zero or one (0 or 1)

Example 11: Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

A is a Boolean matrix but B is not

1.5.1 Boolean matrix operations

i) Join (OR)

- Let A = [a_{ij}] and B = [b_{ij}] be m x n Boolean matrices
- Join of A and B or A $\vee$ B = C = [c_{ij}]

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ij} = 1 \text{ or } b_{ij} = 1 \\ 0 & \text{if } a_{ij} \text{ or } b_{ij} \text{ are both } 0 \end{cases}$$

Example 12: Let A = $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Therefore, A $\vee$ B = $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

ii) Meet (AND)

- Let A = [a_{ij}] and B = [b_{ij}] be m x n Boolean matrices
- The meet of A and B or A $\wedge$ B = D = [d_{ij}]

$$d_{ij} = \begin{cases} 1 & \text{if } a_{ij} \text{ and } b_{ij} \text{ are both 1} \\ 0 & \text{if } a_{ij} = 0 \text{ or } b_{ij} = 0 \end{cases}$$

Example 13: Let A = $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

Therefore, A $\wedge$ B = $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

iii) Boolean Multiplication

- Let A = [a_{ij}] is an m x p Boolean matrix and B = [b_{ij}] is a p x n Boolean matrix. Boolean product of A and B, denoted A ⊙ B, is the m x n Boolean matrix C = [c_{ij}] defined by

$$c_{ij} = \begin{cases} 1 & \text{if } a_{ik} = 1 \text{ and } b_{kj} = 1 \text{ for some k, } 1 \le k \le p \\ 0 & \text{Otherwise} \end{cases}$$

- The multiplication is similar to ordinary matrix multiplication

Example 14: Let A = $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

Therefore, A ⊙ B = $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

- ***Theorem 4*** – basic properties of the Boolean matrix operations

i) (a) $A \vee B = B \vee A$
(b) $A \wedge B = B \wedge A$

ii) (a) $(A \vee B) \vee C = A \vee (B \vee C)$
(b) $(A \wedge B) \wedge C = A \wedge (B \wedge C)$

iii) (a) $A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C)$
(b) $A \vee (B \wedge C) = (A \vee B) \wedge (A \vee C)$

iv) $(A \odot B) \odot C = A \odot (B \odot C)$